GAZİ UNIVERSITY

FACULTY OF ENGINEERING
DEPARTMENT OF INDUSTRIAL ENGINEERING

Lecturer : Dr Ercan Ezin

IE326-DATABASE MANAGEMENT SYSTEMS

WEEK 11: Filters-IN, BETWEEN, LIKE, IS NULL

FILTERS

IN, BETWEEN, LIKE, IS NULL

PostgreSQL IN operator

The `IN` operator allows you to check whether a value matches any value in a list of values.

Here's the basic syntax of the `IN` operator:

```
value IN (value1,value2,...)
```

The `IN` operator returns true if the `value` is equal to any value in the list such as `value1` and `value2`.

The list of values can be a list of literal values including numbers and strings.

PostgreSQL IN operator

In addition to literal values, the IN operator also accepts a list of values returned from a query.

Functionally, the `IN` operator is equivalent to combining multiple boolean expressions with the OR operators:

```
value = value1 OR value = value2 OR ...
```

PostgreSQL IN operator

film
* film_id
title
description
release_year
language_id
rental_duration
rental_rate
length
replacement_cost
rating
last_update
special_features
fulltext

- Using IN operator to retrieve information about the film with id 1, 2, and 3:

SAME

```
SELECT film_id, titleFROM filmWHERE film_id in (1, 2, 3);
```

```
SELECT film_id, title FROM film WHERE film_id = 1 OR film_id = 2 OR film_id = 3;
```

The main difference is , compared to OR operator, IN operator is shorter and execution is fater

- Using IN operator to find the film which have the list of exact values:

```
SELECT title FROM film WHERE title IN ('Alien', 'Contact', 'Arrival') ORDER BY title;
```

- Using OR operator to find the film which have certain words in the name:

```
SELECT title FROM film WHERE title LIKE '%Alien%' OR title LIKE '%Contact%';
```

PostgreSQL IN operator

- The following statement uses the IN operator to find payments whose payment dates are in a list of dates: 2007-02-15 and 2007-02-16:

```
SELECT payment_id, amount, payment_date FROM payment WHERE payment_date::date IN ('2007-02-15', '2007-02-16');
```

In this example, the `payment_date` column has the type `timestamp` that consists of both date and time parts.

To match the values in the `payment_date` column with a list of dates, you need to cast them to date values that have the date part only.

To do that you use the `::` cast operator:

```
payment_date::date
```

For example, if the timestamp value is `2007-02-15 22:25:46.996577`, the cast operator will convert it to `2007-02-15`.

Output

```
payment_id | amount |        payment_date
-----------+--------+----------------------------
      17503 |   7.99 | 2007-02-15 22:25:46.996577
      17504 |   1.99 | 2007-02-16 17:23:14.996577
      17505 |   7.99 | 2007-02-16 22:41:45.996577
      17512 |   4.99 | 2007-02-16 00:10:50.996577
...
```

PostgreSQL NOT IN operator

To negate the `IN` operator, you use the `NOT IN` operator. Here's the basic syntax of the `NOT IN` operator:

```
value NOT IN (value1, value2, ...)
```

The `NOT IN` operator returns `true` if the `value` is not equal to any value in the list such as `value1` and `value2`; otherwise, the `NOT IN` operator returns `false`.

The `NOT IN` operator is equivalent to a combination of multiple boolean expressions with the AND operators:

```
value <> value1 AND value <> value2 AND ...
```

PostgreSQL NOT IN operator

- The following example uses the NOT IN operator to retrieve films whose id is not 1, 2, or 3:

```
SELECT film_id, title FROM film WHERE film_id NOT IN (1, 2, 3) ORDER BY film_id;
```

Output

```
 film_id |            title
---------+----------------------
       4 | Affair Prejudice
       5 | African Egg
       6 | Agent Truman
       7 | Airplane Sierra
       8 | Airport Pollock
```

- The following query retrieves the same set of data but uses the not-equal (<>) and AND operators:

Same output

```
SELECT film_id, title FROM film
WHERE film_id <> 1
AND film_id <> 2
AND film_id <> 3
ORDER BY film_id;
```

PostgreSQL BETWEEN operator

The `BETWEEN` operator allows you to check if a value falls within a range of values.

The basic syntax of the `BETWEEN` operator is as follows:

```
value BETWEEN low AND high;
```

If the `value` is greater than or equal to the `low` value and less than or equal to the `high` value, the `BETWEEN` operator returns `true`; otherwise, it returns `false`.

You can rewrite the `BETWEEN` operator by using the greater than or equal (`>=`) and less than or equal to (`<=`) operators and the logical AND operator:

```
value >= low AND value <= high
```

PostgreSQL BETWEEN operator

If you want to check if a value is outside a specific range, you can use the `NOT BETWEEN` operator as follows:

```
value NOT BETWEEN low AND high
```

The following expression is equivalent to the expression that uses the `NOT BETWEEN` operators:

```
value < low OR value > high
```

In practice, you often use the `BETWEEN` operator in the WHERE clause of the SELECT, INSERT, UPDATE, and DELETE statements.

PostgreSQL BETWEEN operator

payment
* payment_id
customer_id
staff_id
rental_id
amount
payment_date

- The following query uses the BETWEEN operator to retrieve payments with payment_id is between 17503 and 17505:

```
SELECT payment_id, amount FROM payment WHERE payment_id BETWEEN 17503 AND 17505 ORDER BY payment_id;
```

```
 payment_id | amount
------------+--------
      17503 |   7.99
      17504 |   1.99
      17505 |   7.99
```

- The following example uses the NOT BETWEEN operator to find payments with the payment_id not between 17503 and 17505:

```
SELECT payment_id, amount FROM payment WHERE payment_id NOT BETWEEN 17503 AND 17505 ORDER BY payment_id;
```

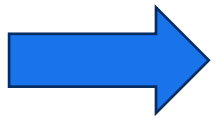
```
 payment_id | amount
------------+--------
      17506 |   2.99
      17507 |   7.99
      17508 |   5.99
      17509 |   5.99
      17510 |   5.99
```

PostgreSQL BETWEEN operator

If you want to check a value against a date range, you use the literal date in ISO 8601 format, which is `YYYY-MM-DD`.

The following example uses the `BETWEEN` operator to find payments whose payment dates are between `2007-02-15` and `2007-02-20` and amount more than 10:

```
SELECT
  customer_id,
  payment_id,
  amount,
  payment_date
FROM
  payment
WHERE
  payment_date BETWEEN '2007-02-15' AND '2007-02-20'
  AND amount > 10
ORDER BY
  payment_date;
```



```
customer_id | payment_id | amount |        payment_date
------------+------------+--------+----------------------------
         33 |      18640 |  10.99 | 2007-02-15 08:14:59.996577
        544 |      18272 |  10.99 | 2007-02-15 16:59:12.996577
        516 |      18175 |  10.99 | 2007-02-16 13:20:28.996577
        572 |      18367 |  10.99 | 2007-02-17 02:33:38.996577
        260 |      19481 |  10.99 | 2007-02-17 16:37:30.996577
        477 |      18035 |  10.99 | 2007-02-18 07:01:49.996577
        221 |      19336 |  10.99 | 2007-02-19 09:18:28.996577
```

PostgreSQL LIKE operator

Suppose that you want to find customers, but you don't remember their names exactly. However, you can recall that their names begin with something like `Jen`.

How do you locate the exact customers from the database? You can identify customers in the `customer` table by examining the first name column to see if any values begin with `Jen`. However, this process can be time-consuming, especially when the `customer` table has a large number of rows.

Fortunately, you can use the PostgreSQL `LIKE` operator to match the first names of customers with a string using the following query:

```
SELECT
  first_name,
  last_name
FROM
  customer
WHERE
  first_name LIKE 'Jen%';
```

Output:

```
first_name | last_name
-----------+----------
 Jennifer  | Davis
 Jennie    | Terry
 Jenny     | Castro
(3 rows)
```

PostgreSQL LIKE operator

The `WHERE` clause in the query contains an expression:

```
first_name LIKE 'Jen%'
```

The expression consists of the `first_name`, the `LIKE` operator and a literal string that contains a percent sign (`%`). The string `'Jen%'` is called a pattern.

The query returns rows whose values in the `first_name` column begin with `Jen` and are followed by any sequence of characters. This technique is called pattern matching.

You construct a pattern by combining literal values with wildcard characters and using the `LIKE` or `NOT LIKE` operator to find the matches.

PostgreSQL LIKE operator

PostgreSQL offers two wildcards:

- Percent sign (`%`) matches any sequence of zero or more characters.
- Underscore sign (`_`) matches any single character.

Here's the basic syntax of the `LIKE` operator:

```
value LIKE pattern
```

The `LIKE` operator returns `true` if the `value` matches the `pattern`. To negate the `LIKE` operator, you use the `NOT` operator as follows:

```
value NOT LIKE pattern
```

The `NOT LIKE` operator returns `true` when the `value` does not match the `pattern`.

If the pattern does not contain any wildcard character, the LIKE operator behaves like the equal (=) operator.

PostgreSQL LIKE operator

The following statement uses the `LIKE` operator with a pattern that doesn't have any wildcard characters:

```
SELECT 'Apple' LIKE 'Apple' AS result;
```

Output:

```
result
--------
 t
(1 row)
```

In this example, the `LIKE` operator behaves like the equal to (`=`) operator. The query returns `true` because '`Apple' = 'Apple'` is `true`.

PostgreSQL LIKE operator

The following example uses the `LIKE` operator to match any string that starts with the letter `A`:

```
SELECT 'Apple' LIKE 'A%' AS result;
```

Output:

```
result
--------
 t
(1 row)
```

The query returns true because the string `'Apple'` starts with the letter `'A'`.

PostgreSQL LIKE operator

customer
* customer_id
store_id
first_name
last_name
email
address_id
activebool
create_date
last_update
active

- The following example uses the LIKE operator to find customers whose first names contain the string er :

```
SELECT first_name, last_name FROM customer WHERE first_name LIKE '%er%' ORDER BY first_name;
```

```
 first_name  |  last_name
-------------+-------------
 Albert      | Crouse
 Alberto     | Henning
 Alexander   | Fennell
```

- The following example uses the LIKE operator with a pattern that contains both the percent (%) and underscore (_) wildcards:

```
SELECT
  first_name,
  last_name
FROM
  customer
WHERE
  first_name LIKE '_her%'
ORDER BY
  first_name;
```

Output:

```
 first_name | last_name
------------+-----------
 Cheryl     | Murphy
 Sherri     | Rhodes
 Sherry     | Marshall
 Theresa    | Watson
(4 rows)
```

The pattern `_her%` matches any strings that satisfy the following conditions:

- The first character can be anything.
- The following characters must be `'her'`.
- There can be any number (including zero) of characters after `'her'`.

PostgreSQL ILIKE operator

PostgreSQL `ILIKE` operator, which is similar to the `LIKE` operator, but allows for **case-insensitive matching**. For example:

```
SELECT
  first_name,
  last_name
FROM
  customer
WHERE
  first_name ILIKE 'BAR%';
```

Output:

```
first_name | last_name
-----------+----------
 Barbara   | Jones
 Barry     | Lovelace
(2 rows)
```

In this example, the BAR% pattern matches any string that begins with BAR, Bar, BaR, and so on. If you use the LIKE operator instead, the query will return no row.

SELECT first_name, last_nameFROM customerWHERE first_name LIKE 'BAR%'; -> NO row returned

PostgreSQL LIKE operator extensions

PostgreSQL also provides some operators that mirror the functionality of `LIKE`, `NOT LIKE`, `ILIKE`, `NOT ILIKE`, as shown in the following table:

Operator	Equivalent
~~	LIKE
~~*	ILIKE
!~~	NOT LIKE
!~~*	NOT ILIKE

```
first_name | last_name
-----------+-----------
 Darlene   | Rose
 Darrell   | Power
 Darren    | Windham
 Darryl    | Ashcraft
 Daryl     | Larue
(5 rows)
```

For example, the following statement uses the ~~ operator to find a customer whose first names start with the string Dar:

```
SELECT first_name, last_name FROM customer WHERE first_name ~~ 'Dar%' ORDER BY first_name;
```

PostgreSQL LIKE operator

Sometimes, the data, that you want to match, contains the wildcard characters `%` and `_`. For example:

```
The rents are now 10% higher than last month
The new film will have _ in the title
```

To instruct the `LIKE` operator to treat the wildcard characters `%` and `_` as regular literal characters, you can use the `ESCAPE` option in the `LIKE` operator:

```
string LIKE pattern ESCAPE escape_character;
```

PostgreSQL LIKE operator

The following statement uses the `LIKE` operator with the `ESCAPE` option to treat the `%` followed by the number `10` as a regular character:

```
SELECT * FROM t
WHERE message LIKE '%10$%%' ESCAPE '$';
```

Data in message column

```
message
-------------------------------------------
 The rents are now 10% higher than last month
 The new film will have _ in the title
(2 rows)
```

Output:

```
message
-------------------------------------------
 The rents are now 10% higher than last month
(1 row)
```

In the pattern `%10$%%`, the first and last `%` are the wildcard characters whereas the `%` appears after the escape character `$` is a regular character.

Example- Insert the data and answer Q1 & Q2

- CREATE TABLE employee_directory (emp_id SERIAL PRIMARY KEY, first_name VARCHAR(50), last_name VARCHAR(50), job_title VARCHAR(100), department_code VARCHAR(10), salary NUMERIC(10, 2));

- INSERT INTO employee_directory (first_name, last_name, job_title, department_code, salary) VALUES ('alice', 'SMITH', 'Lead Software Engineer', 'IT', 95000), ('Bob', 'Smith', 'junior software engineer', 'IT', 55000), ('ALICIA', 'Vance', 'DATABASE ADMINISTRATOR', 'DB', 88000), ('Charlie', 'Smithe', 'Project Manager', 'MGMT', 75000), ('David', 'Miller', 'Senior Engineer', 'IT', 105000), ('Eve', 'Adamson', 'Junior Analyst', 'FIN', 48000);

Q1: Management needs a list of all employees whose **last name** is exactly 'Smith'. Write a query that retrieves their first_name and last_name using the LIKE operator. Observe how many results you get. Then, write a second query using ILIKE to find all variations (e.g., SMITH, Smith, smith). Concatenate the names into a single column named full_employee_name.

Q2: Retrieve all columns for employees who have the word 'engineer' anywhere in their `job_title` (regardless of capitalization). Filter the results so that only those with a `salary` greater than 60,000 are shown and sort the final list by `salary` in descending order.

PostgreSQL NULL operator

In the database world, NULL means missing information or not applicable. NULL is not a value, therefore, you cannot compare it with other values like numbers or strings.

The comparison of NULL with a value will always result in NULL. Additionally, NULL is not equal to NULL so the following expression returns NULL:

```
SELECT null = null AS result;
```

Output:

```
result
--------
 null
(1 row)
```

PostgreSQL IS NULL operator

To check if a value is NULL or not, you cannot use the equal to (`=`) or not equal to (`<>`) operators. Instead, you use the `IS NULL` operator.

Here's the basic syntax of the `IS NULL` operator:

```
value IS NULL
```

The `IS NULL` operator returns true if the `value` is NULL or false otherwise.

To negate the `IS NULL` operator, you use the `IS NOT NULL` operator:

```
value IS NOT NULL
```

The `IS NOT NULL` operator returns true if the value is not NULL or false otherwise.

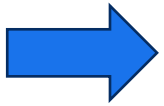
PostgreSQL IS NULL operator examples

- Please note that the psql program displays NULL as an empty string by default. To change how psql shows NULL in the terminal, you can use the command: \pset null null. It will display NULL as null.

address
* address_id
address
address2
district
city_id
postal_code
phone
last_update

The following example uses the `IS NULL` operator to find the addresses from the `address` table that the `address2` column contains `NULL`:

```
SELECT
  address,
  address2
FROM
  address
WHERE
  address2 IS NULL;
```



Output:

```
address         | address2
----------------------+----------
 47 MySakila Drive    | null
 28 MySQL Boulevard   | null
 23 Workhaven Lane    | null
 1411 Lillydale Drive | null
(4 rows)
```

PostgreSQL IS NOT NULL operator example

address
* address_id
address
address2
district
city_id
postal_code
phone
last_update

The following example uses the `IS NOT NULL` operator to retrieve the address that has the `address2` not NULL:

```
SELECT
  address,
  address2
FROM
  address
WHERE
  address2 IS NOT NULL;
```

Output:

```
address                  | address2
-----------------------------------+----------
 1913 Hanoi Way                    |
 1121 Loja Avenue                  |
 692 Joliet Street                 |
 1566 Inegl Manor                  |
```

- Notice that the address2 is empty, not NULL. This is a good example of bad practice when it comes to storing empty strings and NULL in the same column.
- To fix it, you can use the UPDATE statement to change the empty strings to NULL in the address2 column. WE will see it later.

DATA FOR QUESTIONS

Create following table and insert the data to answer questions in the following slide:

- --WEEK 9 & 10: Structural Setup

CREATE TABLE logistics_hubs (hub_id SERIAL PRIMARY KEY, hub_name VARCHAR(100) NOT NULL, city VARCHAR(50), region VARCHAR(20), capacity_teu INT, manager_note TEXT, operating_cost NUMERIC(12, 2));

- -- WEEK 11: Data Insertion

INSERT INTO logistics_hubs (hub_name, city, region, capacity_teu, manager_note, operating_cost) VALUES ('North Star Terminal', 'Istanbul', 'Marmara', 5000, 'Primary export gateway.', 125000.00), ('Aegean Gate', 'Izmir', 'Aegean', 3500, NULL, 85000.50), ('Capital Logistics', 'Ankara', 'Central', 2000, 'High-security inland hub.', 92000.00), ('South Port A1', 'Antalya', 'Mediterranean', 1500, 'Seasonal congestion expected.', 45000.00), ('South Port B2', 'Mersin', 'Mediterranean', 4200, '', 110000.00), ('Eastern Link', 'Erzurum', 'East', 800, 'Cold weather protocols active.', 32000.00);

QUESTIONS

1. Retrieve the hub_name and capacity_teu for all hubs located in the 'Marmara', 'Aegean', or 'Central' regions. Sort the results by capacity_teu in descending order.
2. Find the hub_name and city of any hub whose name contains the word 'Port' (case-insensitive) and has an operating_cost between 40,000 and 120,000.
3. List all columns for hubs where the manager_note is either missing (NULL) or was recorded as an empty string.
4. Retrieve the top 2 hubs with the highest capacity_teu that are located in the 'Mediterranean' region. Use the SQL-standard FETCH clause instead of LIMIT.

THE END

CHECK OUT COURSE WEBSITE: HTTPS://GAZI-END-MUH.GITHUB.IO

GOT ANY QUESTIONS LATER?

SEND ME AN EMAIL: DR.ERCAN.EZIN@GMAIL.COM